



Assembly Instructions for Parker Fittings

Release 07/2001

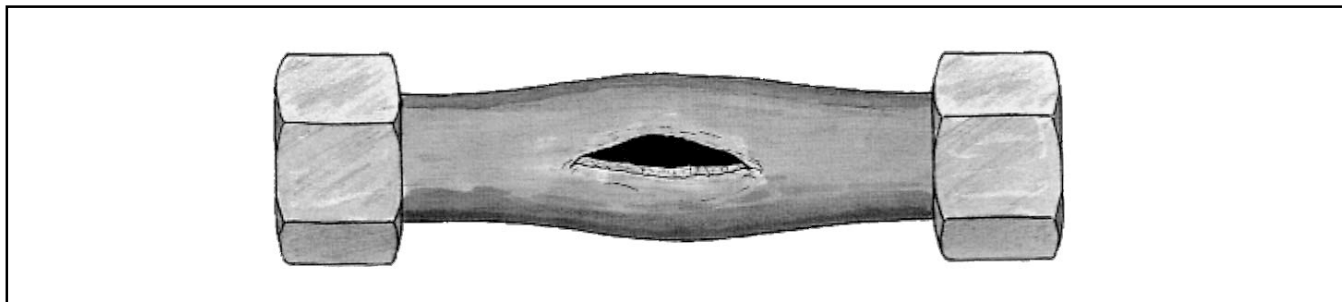
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Safety Instructions

EO Tube Fittings are Safe High-Pressure Connections.



A carefully assembled EO fitting will provide a sealed joint even up to tube burst. Experience has shown that break-downs, tightening-up and leaks can be avoided by following the safety instructions. Please review your fitting procedures.

Safety Instructions Assembly

- During a progressive ring and EO-2 fitting assembly the tube has to bottom up in the stud or in the tool. Without tube bottoming the ring cannot bite sufficiently. Under load the connection can fail due to tube shear.
- Uncompletely assembly will reduce the pressure and vibration capability of a fitting. It can reduce the life cycle time of a connection and leakage can occur. In extreme cases the connection can fail due to tube shear or tube crack. Presetted progressive ring fittings need a final assembly according to assembly instructions.
- After opening a tube connection, the unit has to be retighten with the same force used during prior assembly. Undertightening can result in leakage and can reduce the vibration capability. Overtightening can reduce the possibilities of repeated assembly. In extreme cases the components can be destroyed.
- Stainless steel progressive ring fittings have to be presetted in hardened tools. Otherwise the connection may fail under load due to tube shear.

General safety instructions

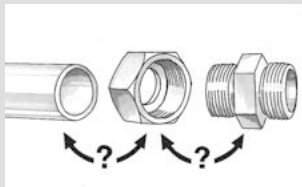
- EO tube fittings are intended solely for connections for fluid applications.
- Observe tube recommendations. Non-standard materials or tolerances lead to incorrect fitting.
- Do not use bearing balls, fitting pins or tapered pins instead of EO plugs (VKA) as blanking parts for 24° cones.
- Do not assemble progressive rings and functional nuts on selfmade standpipe stud ends. There is a risk of false assembly with the result of connection shear under load.
- The use of steel cutting rings for stainless steel tubes or other unauthorised tool combinations leads to incorrect assembly.
- Tube connection and fitting body once assembled, should remain together. Fitting body is to be used once only for pre-assembly.
- Air bleeding of tube fittings which are under pressure can be dangerous.
- Tube under tension can lead to vibration failure. Tube length and bend angles are to be adhered to precisely. Fix tube lines with tube clamps.

- Tubes are not to be clamped to one another but to suitable fixed points. Plate brackets, cable connections and fixing elements are not suitable. Tubes are not mountings on which to integrate other components e.g. filters, ventilators or shut-off valves.
- Prevent oscillation, pressure surges and inherent strain by using flexible hoses for example.
- Under and overtightening of fittings during assembly reduces the capacity for withstanding pressure and vibration loads and therefore reduces the life of the tube fitting. Leaks from the tube can occur under these circumstances.
- When dismantling/transporting and re-assembling, make sure that no dirt enters the system, that the connection elements (threads, sealing surfaces) are not damaged, seals are not lost and tubes are not bent or flattened. We recommend the use of suitable protective caps.
- Disassembled fittings are to be checked for accuracy and damage and replaced if necessary.
- Do not use hand cutters or tube cutters.
- Impurities and swarf can lead to disruptions in the system and leaks.
- The operating parameters given (e.g. pressure, temperature, medium compatibility) are to be adhered to.
- Avoid flow rates > 8 m/s. The resulting forces are high and can destroy the tube lines.
- Relevant guidelines (e.g. CE, ISO, BG, TÜV, DIN) are to be observed.
- Weld fittings are manufactured out of weldable materials. No other fittings are suitable for welding.
- EO Niromont is a high-performance lubricant. The use of other lubricants generally leads to an increase in assembly force.
- The tools and lubricants recommended by EO guarantee safe assembly.
- If tools and assembly equipment of other manufacturers are used, the function and assembly results have to be checked.
- Fittings are to be handled with care.
- Tubelines need to be adapted tension free of the relevant connectors before assembly. An easy turning of the nut is required for the complete thread length. Otherwise leakage can occur. In extreme cases with additional vibrations tube cracks can occur.
- Vibrations have to be clamped by tube clamps. Independent vibration units need to be separated with hoses. Otherwise tube cracks can occur.

In case of doubt please contact your Parker representative!

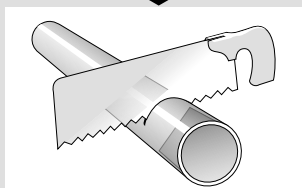
General

Assembly of EO-2 and Progressive Ring fitting is always following the same pattern:



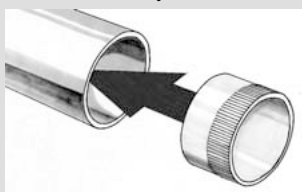
Material combinations

- Select suitable components according to tube material



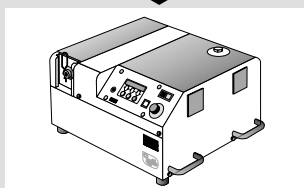
Tube preparation

- Cut and deburr thoroughly
- Do not assemble under tension
- Clamp onto rigid fixtures



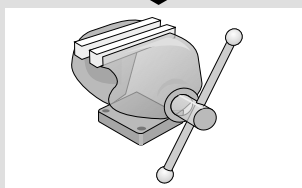
Support sleeves

- Apply support sleeves when necessary



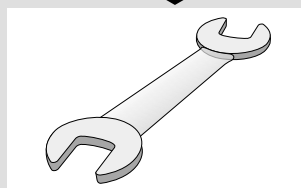
EOMAT/ EO-KARRYMAT

- Preferred and
- Most efficient method



Assembly in vice

- Reliable method
- Only economical for assembly of small quantities



Direct assembly

- Simple procedure for single assemblies of small dimensions
 - Not economic for series assemblies
- Tubes $\varnothing$ 30, 35, 38 and 42 mm must be pre-assembled in vice
- Stainless steel EO Progressive rings must be preassembled using a hardened tool



Assembly check

- Check assembly result
- Incorrect assemblies must be corrected or scraped

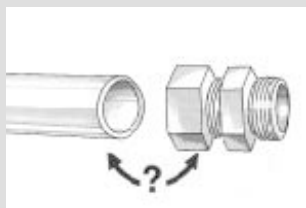


Installation

- Final fitting assembly according to instruction



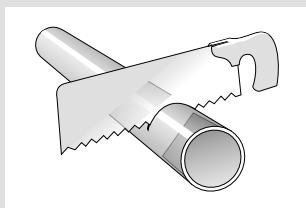
EO-Progressive Ring Assembly Instructions



Material Combinations

- Select suitable EO Progressive Ring Fitting

Tube = Material	EO-Progressive Ring Fitting	Assembly Instructions
Steel	Steel	
Stainless Steel	Stainless Steel	Pre-assembly with a hardened tool required
Copper	Brass	
Plastic e.g. Polyamid	Steel	Support sleeve E required

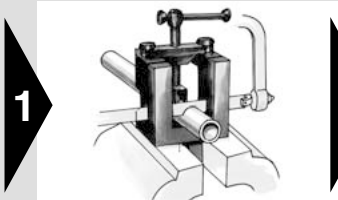


Tube preparation

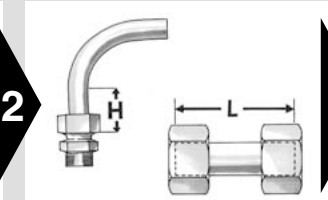
- Cut and deburr thoroughly
- Do not assemble under tension
- Clamp onto rigid fixtures

Minimum length of straight tube ends

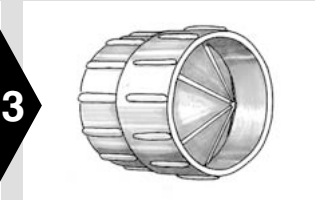
Series	L											S										
Tube O.D.	06	08	10	12	15	18	22	28	35	42		06	08	10	12	14	16	20	25	30	38	
H min	31	31	33	33	36	38	42	42	48	48		35	35	37	37	43	43	50	54	58	65	
L min	39	39	42	42	45	48	53	53	60	60		44	44	47	47	54	54	63	68	73	82	



- Cut tube squarely
- max $\pm 1^\circ$ deviation
- ⚠ Do not use pipe cutters
- EO tube-cutting tool (AV)



- Minimum lengths of straight tube ends (see above table)

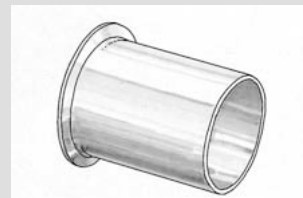


- Remove internal and external burrs
- Chamfer up to $0.3 \times 45^\circ$ permissible
- Recommendation: In-Ex Tube Deburring Tool 226



Support sleeves VH

- Support sleeve VH for thin wall or soft metal tubes
- See catalogue assembly instruction booklet for VH application



Tube insert E

- Support sleeve E for plastic tubes



- Insert support sleeve like shown



- Drive VH into tube-end

VH selection chart for EO-Progressive Ring

For steel tubes material ST 37.4
and for stainless steel tubes material 1.4571 and 1.4541

Wall thickness	4	5	6	8	10	12	14	15	16	18	20	22	25	28	30	35	38	42
3																		
2.5																		
2																		
1.5																		
1																		
0.75																		

For soft metal tubes (e. g. copper)

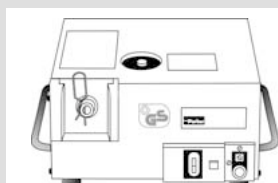
Wall thickness	4	5	6	8	10	12	14	15	16	18	20	22	25	28	30	35	38	42
3.5																		
3																		
2.5																		
2																		
1.5																		
1																		
0.75																		
0.5																		

- Support sleeve required.
- Support sleeve required for heavily loaded lines (vibrations).

EO-Progressive Ring Assembly Instructions

Preassembly with EOMAT/EO-KARRYMAT

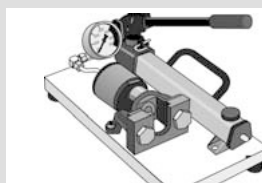
- Preferred and
- Most efficient Method



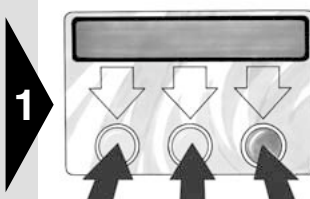
EOMAT II



EOMAT III/A



EO-KARRYMAT



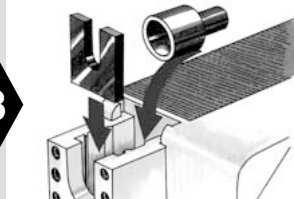
1

- EOMAT II: Adjustment according to pressure chart on machine
- EOMAT III/A: Menu selection
- EO-KARRYMAT: Refer to chart on machine
- Non-EOMAT-machines: check suitability



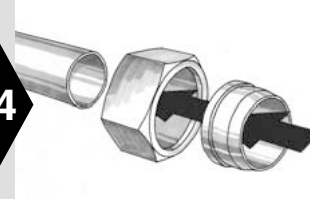
2

- The assembly cones (MOK) must be checked regularly (after 50 pre-assemblies) with cone gauges (KONU)
- The rear of the gauge must protrude slightly above the top face of the cone or may be flush
- Check insertion depth p. B16 (see p. B17)



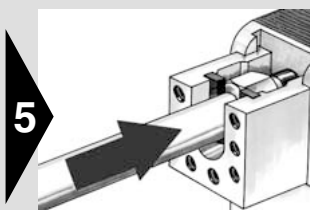
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- Insert proper tools
- Clean and lubricate assembly cones regularly
- EO-KARRYMAT: Close valve on handpump
- 2-piece backing plates for 35-L and 42-L



4

- Slide nut and progressive ring as shown onto the end of the tube



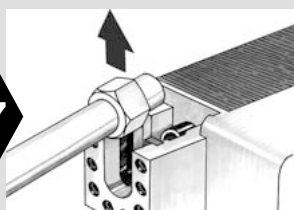
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- Place tube with progressive ring and nut into the die
- Press tube-end firmly into the assembly cone



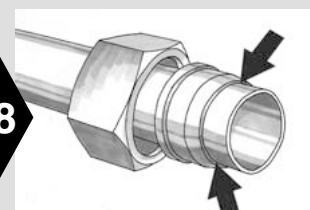
6

- Hold tube firmly
- EOMAT: Press and hold start button
- Use support and foot switch for long tubes
- EO-KARRYMAT: Operate handpump until assembly pressure is reached



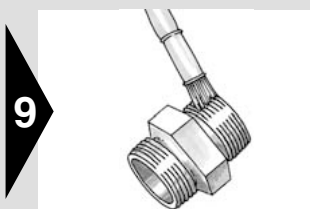
7

- After completion of pre-assembly, remove the tube for assembly check
- EO-KARRYMAT: Open valve on handpump



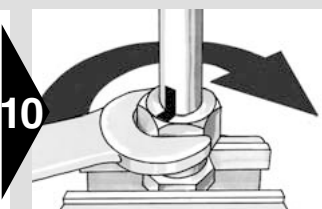
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- ⚠ Check to make sure that a visible collar covers the front of the first cutting edge
- It does not matter if the ring can be rotated on the tube end



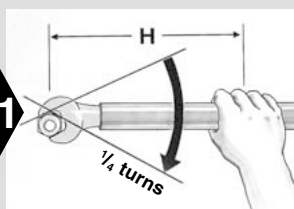
9

- ⚠ Threads of stainless steel fittings must be lubricated
- EO-NIROMONT is a special high-performance lubricant for stainless steel fittings



10

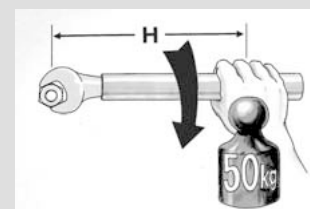
- Assemble fitting until wrench-tight (without spanner extension)
- ⚠ Mark position of nut



11

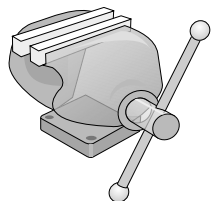
- ⚠ Then tighten fitting firmly by 1/4 turns (1 1/2 flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. chart C
- ⚠ Threads on stainless steel must be lubricated

Chart C



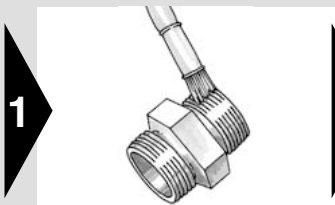
Size	Spanner length H [mm]
18-L 16-S	300
22-L	400
28-L 20-S	500
35-L 25-S	900
42-L 30-S	1200
38-S	1500

EO-Progressive Ring Assembly Instructions



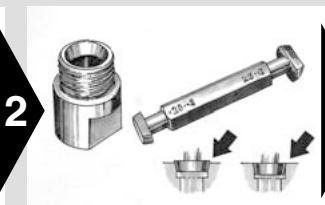
Pre-Assembly in vice with VOMO

- Reliable method
 - Only economical for assembly of small quantities
- Stainless steel EO Progressive rings must be preassembled using a hardened tool



1

- ⚠ For stainless steel assembly, threads must be lubricated.
- EO-NIROMONT is a special high-performance lubricant for stainless steel fittings



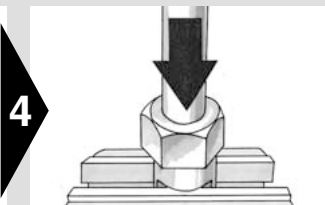
2

- Cones of pre-assembly bodies must be checked regularly (after 50 pre-assemblies) with cone gauges (KONU)
- ⚠ The rear of the gauge must protrude slightly above the top face of the cone or may be flush
- Check insertion depth (see p. B16)



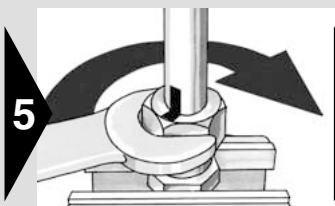
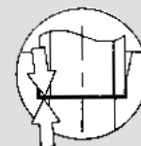
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- Use preassembly tool VOMO
- Fitting body may be used one time only (not for stainless steel)
- Screw on nut until finger-tight



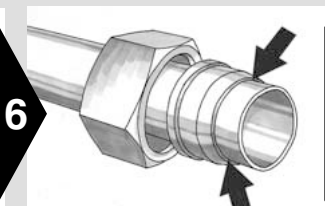
4

- ⚠ Press tube-end firmly into the assembly cone



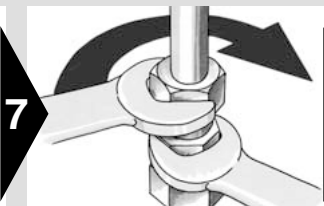
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- ⚠ Mark position of the nut
- Tighten the nut by 1 1/4 turns
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)



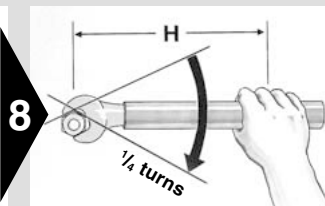
6

- Assembly check:**
- Loosen nut
- ⚠ Check to make sure that a visible collar covers the front of the first cutting edge
- ⚠ It does not matter if the ring can be rotated on the tube end



7

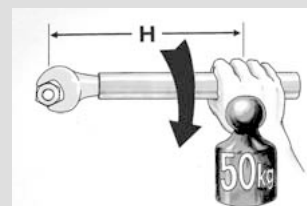
- Assemble fitting until wrench-tight (without spanner extension)
- ⚠ Mark position of nut



8

- ⚠ Then tighten fitting firmly by 1/4 turns (1 1/2 flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

Chart C



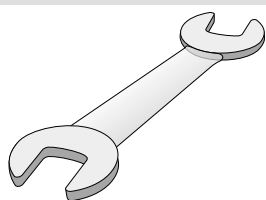
Size	Spanner length H [mm]
18-L 16-S	300
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28-L 25-S	500
35-L 30-S	900
42-L 38-S	1200
	1500

EO-Progressive Ring Assembly Instructions

Direct Assembly

- Simple procedure for single assemblies of small dimensions
- Not economic for series assemblies

- ⚠ Tubes Ø 30, 35, 38 and 42 mm must be preassembled in vice
- ⚠ Stainless steel connections have to be assembled using pre-assembly tool



1

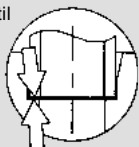


- ⚠ Lubrication of threads will reduce wear and assembly forces.
- ⚠ Threads on stainless steel fittings must be lubricated
- ⚠ EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

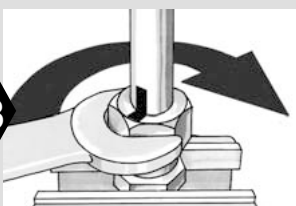
2



- Screw on nut until finger-tight
- Press tube-end firmly into fitting body

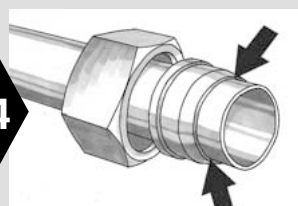


3



- Mark position of the nut
- Tighten the nut by 1½ turns
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

4



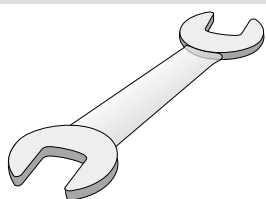
Assembly check:

- Loosen nut
- Check to make sure that a visible collar covers the front of the first cutting edge
- It does not matter if the ring can be rotated on the tube end

Repeated Assembly

- Each time the tube-end has been disconnected, the fitting must be properly tightened again

- ⚠ EO Progressive Rings cannot be replaced, once assembled

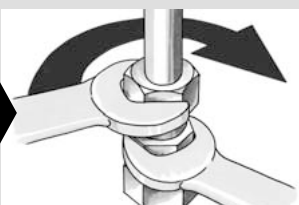


1



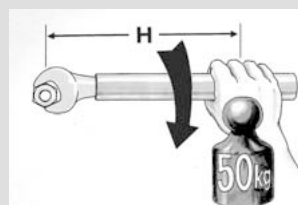
- ⚠ Threads on stainless steel fittings must be lubricated
- ⚠ EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

2



- Each time the fitting has been loosened, re-assembly must be performed with the same torque as initial assembly
- The body must be held rigid
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

Chart C



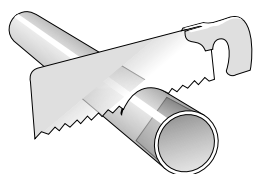
Size		Spanner length H [mm]
18-L	16-S	300
22-L		400
28-L	20-S	500
35-L	25-S	900
42-L	30-S	1200
	38-S	1500



EO-2 Assembly Instructions

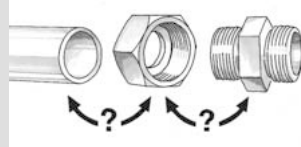
Detailed Assembly-instructions are included in each EO-2 product box.

Details on EOMAT setting and selection of support sleeves can be found there as well.



Tube preparation

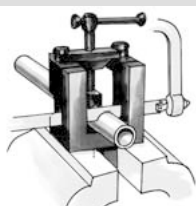
- Cut and deburr thoroughly
- Do not assemble under tension
- Clamp onto rigid fixtures



Material combinations

- Select suitable FM-type

1



- Cut tube squarely
- max $\pm 1^\circ$ deviation
- ⚠ Do not use pipe cutters
- EO tube-cutting tool (AV)

2

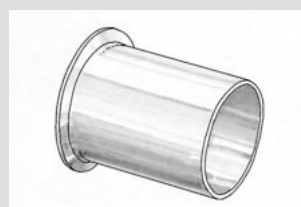


- Remove internal and external burrs
- Chamfer up to $0.3 \times 45^\circ$ permissible
- Seal can be damaged by large burrs

	Steel tube	Stainless steel tube	Plastic tube
Steel fitting	FM...A3C	FM...SSA	FM...A3C+E
Stainless steel fitting	—	FM...71	FM...71+E

Support sleeves VH

- Support sleeve VH for thin wall or soft metal tubes



Tube insert E

- Tube insert E for plastic tubes

1



- Support-sleeve selection: see instruction shipped with product

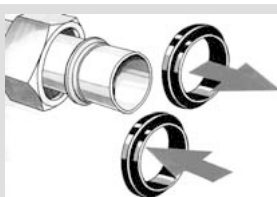
2



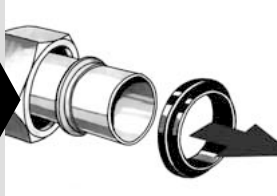
- Drive VH into tube-end

Replacement of sealing ring/Repeated assembly

- Sealing ring DOZ can be changed separately

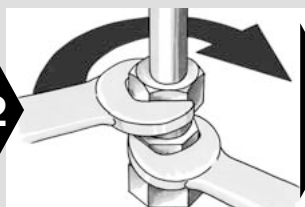


1



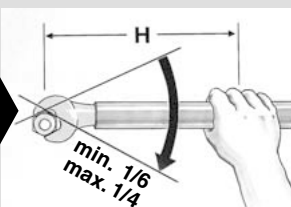
- After disassembly, sealing ring can be pulled off the tube-end
- Check for damage and replace if necessary
- Abrasion on outer rubber parts does not effect performance

2



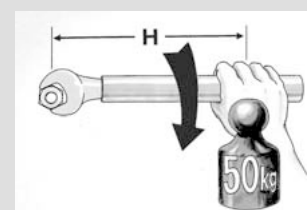
- Assemble fitting until wrench-tight (without spanner extension)

3



- ⚠ Then tighten fitting firmly by min $\frac{1}{6}$ (max $\frac{1}{4}$) turns (1 to $1\frac{1}{2}$ flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

Chart C

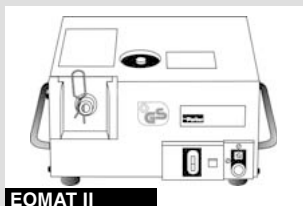


Size	Spanner length H [mm]
18-L 16-S	300
22-L 20-S	400
28-L 25-S	500
35-L 30-S	900
42-L 38-S	1200
	1500

EO-2 Assembly Instructions

Assembly with EOMAT/EO-KARRYMAT

- Preferred and
- Most efficient Method



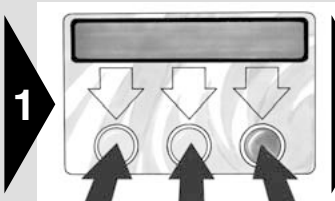
EOMAT II



EOMAT III/A



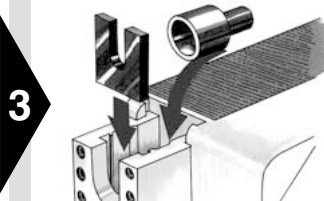
EO-KARRYMAT



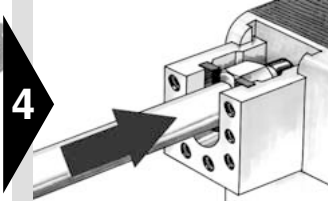
- EOMAT II: Adjustment according to pressure **chart A** on machine (see instructions shipped with product box)
- EOMAT III/A: Menu selection
- EO-KARRYMAT: Refer to chart on machine
- Non-EOMAT-machines: check suitability



- Check insertion depth (see p. B17)



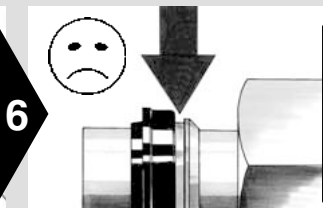
- Insert proper tools
- 2-piece tube backing plates for 35-L and 42-L
- EO-KARRYMAT: Close valve on handpump



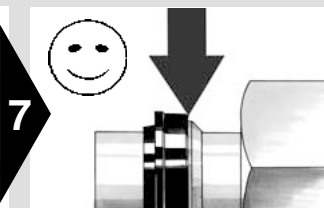
- Place tube with functional nut into the die
- Press tube end firmly into the assembly cone



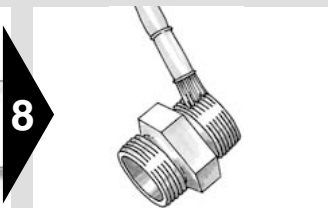
- Hold tube firmly
- EOMAT: Press and hold start button
- Use support and foot switch for long tubes
- EO-KARRYMAT: Operate handpump until assembly pressure is reached. Then open valve on handpump



- ⚠ **Gap not closed:** Check all components, tube, machine, tools and pressure setting. Repeat assembly with increased pressure if necessary



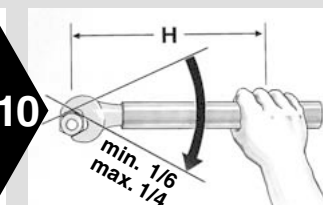
- **Assembly check:** Gap between sealing ring and retaining ring must be closed
- Little relaxation (max 0.2 mm) is allowed



- ⚠ Threads of stainless steel fittings must be lubricated
- ⚠ EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

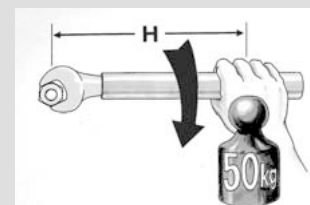


- Assemble fitting until wrench-tight (without spanner extension)



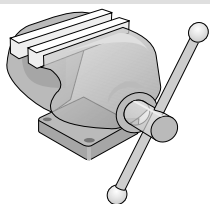
- ⚠ Then tighten fitting firmly by min $\frac{1}{6}$ (max. $\frac{1}{4}$) turns (1 to $1\frac{1}{2}$ flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (**chart C**)

Chart C



Size	Spanner length H [mm]
18-L 16-S	300
22-L 20-S	400
28-L 20-S	500
35-L 25-S	900
42-L 30-S	1200
38-S	1500

EO-2 Assembly Instructions



Assembly in vice

- Reliable method
- Only economical for assembly of small quantities

1



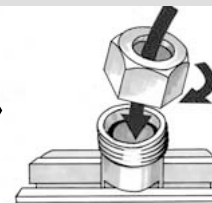
- ⚠ Threads on stainless steel fittings must be lubricated
- ⚠ EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

2



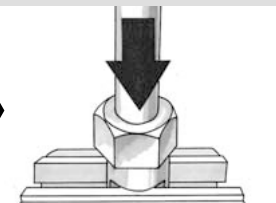
- Check insertion depth **Chart B** (see p. B17)

3

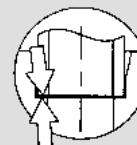


- Use preassembly tool VOMO
- Fitting body may be used one time only
- Screw on nut until finger-tight

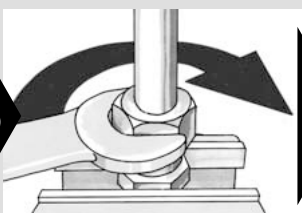
4



- ⚠ Press tube-end firmly into the assembly cone

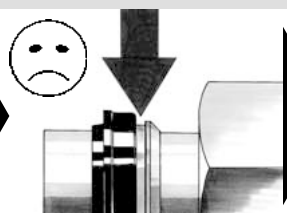


5



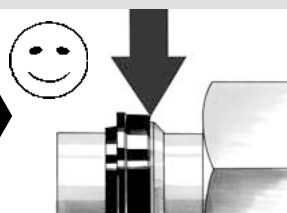
- Tighten until sharp increase of resistance (approx. 1 to 1 1/2 turns)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (**Chart C**)

6



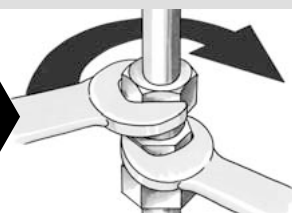
- ⚠ **Gap not closed:** Check all components including tube

7



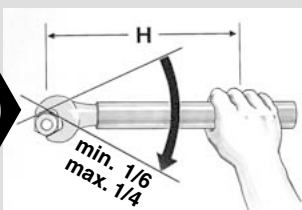
- **Assembly check:** Gap between sealing ring and retaining ring must be closed
- Little relaxation (max 0.2 mm) is allowed

8



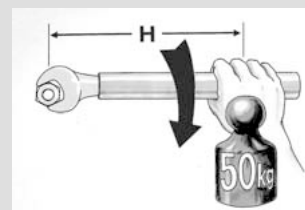
- Assemble fitting until wrench-tight (without spanner extension)

9



- ⚠ Then tighten fitting firmly by min 1/6 (max. 1/4) turns (1 to 1 1/2 flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (**Chart C**)

Chart C

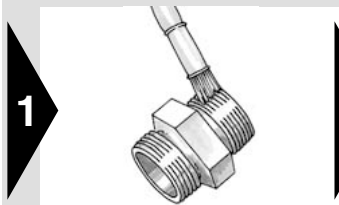
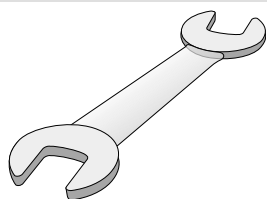


Size		Spanner length H [mm]
18-L	16-S	300
22-L		400
28-L	20-S	500
35-L	25-S	900
42-L	30-S	1200
	38-S	1500

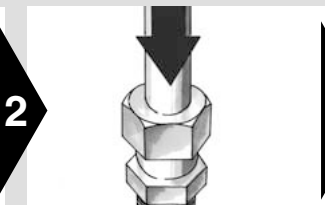
EO-2 Assembly Instructions

Direct Assembly

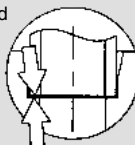
- Simple procedure for single assemblies of small dimensions
 - Not economic for series assemblies
- ⚠ Tubes Ø 30, 35, 38 and 42 mm must be preassembled in vice



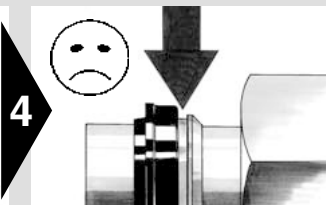
- ⚠ Threads on stainless steel fittings must be lubricated
- ⚠ EO-NIROMONT is a special high-performance lubricant for stainless steel fittings



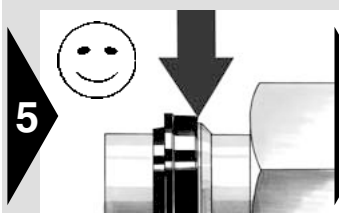
- ⚠ Press tube-end firmly into the assembly cone



- Tighten until sharp increase of resistance (approx. 1 to 1 1/2 turns)
- Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)



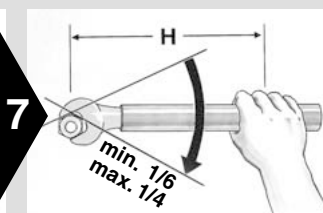
- ⚠ **Gap not closed:** Check all components including tube



- **Assembly check:** Gap between sealing ring and retaining ring must be closed
- Little relaxation (max 0.2 mm) is allowed

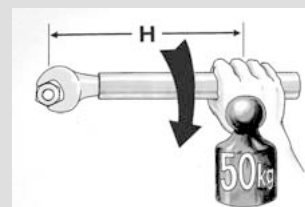


- Assemble fitting until wrench-tight (without spanner extension)



- ⚠ Then tighten fitting firmly by min 1/6 (max 1/4) turns (1 to 1 1/2 flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

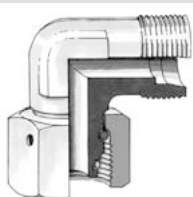
Chart C



Size	Spanner length H [mm]
18-L 16-S	300
22-L	400
28-L 20-S	500
35-L 25-S	900
42-L 30-S	1200
38-S	1500



Swivel Nut and Weld Fittings

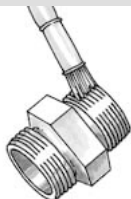


Assembly of Swivel Nut and Weld Fittings

(e. g. EW, ET, EL, EGE, RED, VKA, SKA)

- Final assembly of swivel nut fittings must be made in appropriate fittings

1



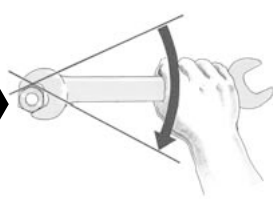
- ⚠ Threads of stainless steel fittings must be lubricated
- EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

2



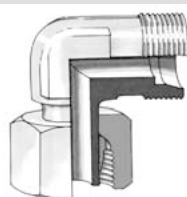
- Screw on nut by hand until handtight

3



- ⚠ Then tighten fitting firmly by 1/4 turns (1 1/2 flats)

Standpipe fittings

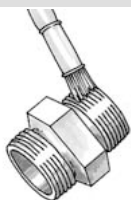


Final Assembly of Factory Pre-assembled Standpipe Fittings

(e. g. EVW, EVT, EVL, EVGE, KOR)

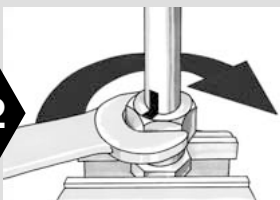
- For all fittings delivered pre-assembled from the factory the final assembly is performed in the appropriate fitting body

1



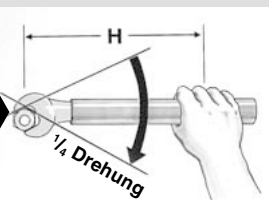
- ⚠ Threads of stainless steel fittings must be lubricated
- EO-NIROMONT is a special high-performance lubricant for stainless steel fittings

2



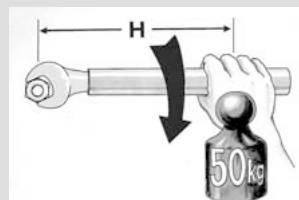
- Assemble fitting until wrench tight (without spanner extension)
- ⚠ Mark position of nut

3



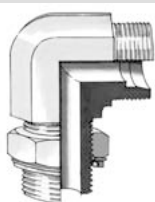
- ⚠ Then tighten fitting firmly by 1/4 turns (1 1/2 flats)
- ⚠ Recommended to use spanner extension for sizes over 20 mm O.D. (Chart C)

Chart C



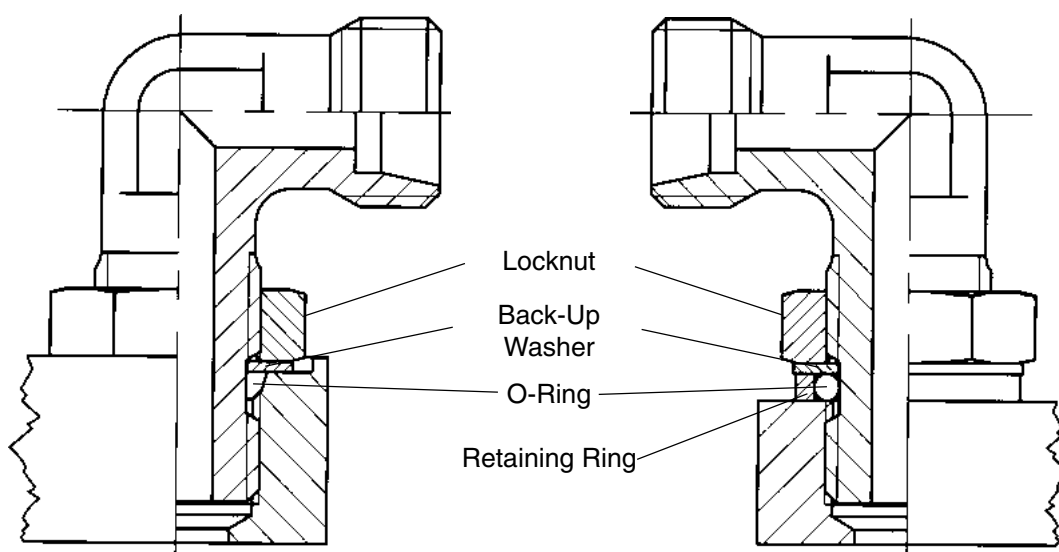
Size		Spanner length H [mm]
18-L	16-S	300
22-L		400
28-L	20-S	500
35-L	25-S	900
42-L	30-S	1200
	38-S	1500

Adjustable Fittings with Locknut



Assembly of the orientable joint (e. g. WEE, VEE, TEE, LEE)

⚠ Assembly steps must be done in right order



● Fitting without Retaining Ring for ISO 6149 or UN/UNF ports

● Fitting with Retaining Ring for BSPP or Metric Parallel ports with wide or *small* spot faces

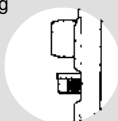
1



- Screw back locknut as far as possible
- O-ring and back-up washer in the non-threaded section should be placed nearest to the locknut



- Lubricate the O-ring
- With BSPP and metric parallel version slip retaining ring over the O-ring



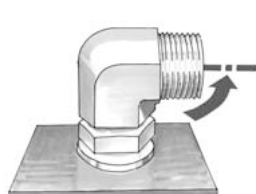
2



- Screw the fitting in the port by hand until retaining-ring or back-up washers bottoms



3

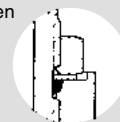


- To adjust direction, turn back to a maximum of one full turn

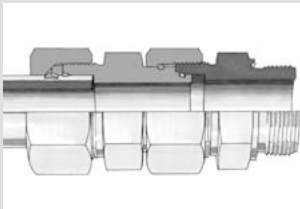
4



- Hold body in desired position and tighten locknut

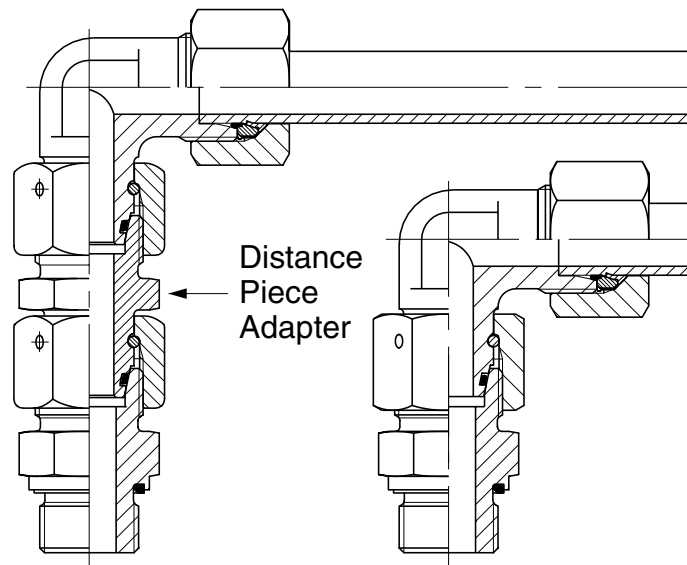


Replacement of a cutting ring connection

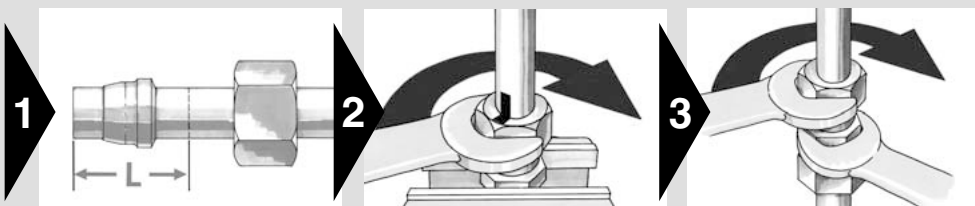


Distance Piece Adapter DA

- EO Distance piece adapters allows to replace Bite-type connections on existing pipeworks easily or to retrofit using EO-2
- The existing tubes can be re-used



- Use as an extension for stacked assemblies



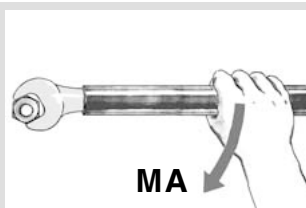
- Shorten tube end by distance "L" (for "L" dimension see page C172)
- Remove obsolete nut

- Assemble new EO-2 functional nut or nut with EO progressive ring

- Screw and tighten distance piece adapter on tube end



Tightening Torques (MA) for Port Connections



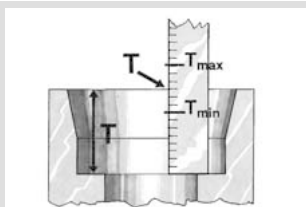
- acc. to ISO 9974-1 / ISO 6149-1 / DIN 3852-T1-form X / DIN 3852-T3-form W
- Metric ISO thread M

Series	Tube O.D.	Thread G	MA / Nm								
			Straight male stud fittings with port tapping				Non-return valves RHV/RHZ	Banjo fittings		Blanking plugs	
			Form A for sealing washer	Form B with face	Form E with ED-sealing	Form F with O-Ring-sealing	Form E with ED-sealing	WH / TH	SWVE	VSTI-ED Form E with ED-sealing	VSTI-OR Form F with O-Ring-sealing
L	6	M10 x 1.0	9	18	18	15	18	18	18	12	20
	8	M12 x 1.5	20	30	25	25	25	45	35	25	
	10	M14 x 1.5	35	45	45	35	35	55	50	35	
	12	M16 x 1.5	45	65	55	40	50	80	60	55	
	15	M18 x 1.5	55	80	70	45	70	100	80	65	
	18	M22 x 1.5	65	140	125	60	125	140	120	90	
	22	M27 x 2.0	90	190	180	100	145	320	130	135	
	28	M33 x 2.0	150	340	310	160	210	360		225	
	35	M42 x 2.0	240	500	450	210	360	540		360	
	42	M48 x 2.0	290	630	540	260	540	700		360	
S	6	M12 x 1.5	20	35	35	35	35	45	35		35
	8	M14 x 1.5	35	55	55	45	45	55	50		45
	10	M16 x 1.5	45	70	70	55	55	80	60		55
	12	M18 x 1.5	55	110	90	70	70	100	80		70
	14	M20 x 1.5	55	150	125	80	100	125	110	80	80
	16	M22 x 1.5	65	170	135	100	125	135	120		100
	20	M27 x 2.0	90	270	180	170	135	320	135		170
	25	M33 x 2.0	150	410	310	310	210	360			310
	30	M42 x 2.0	240	540	450	330	360	540			330
	38	M48 x 2.0	290	700	540	420	540	700			420

Tolerance of tightening torques listed in above table: + 10%

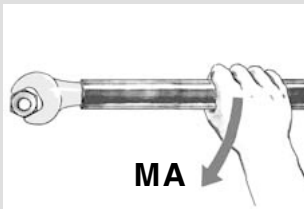
Note: Lubricate stud with hydraulic oil before screwing in! Tightening torques relate to counterpart made of steel.

Checklist “inside depth” for presetting tool (MOK and VOMO)



Type	T _{min}	T _{max}	Type	T _{min}	T _{max}
6-L	6.95	7.05	6-S	6.95	7.05
8-L	6.95	7.05	8-S	6.95	7.05
10-L	6.95	7.05	10-S	7.45	7.55
12-L	6.95	7.05	12-S	7.45	7.55
15-L	6.95	7.05	14-S	7.95	8.05
18-L	7.45	7.55	16-S	8.45	8.55
22-L	7.45	7.55	20-S	10.45	10.55
28-L	7.45	7.55	25-S	11.95	12.05
35-L	10.45	10.55	30-S	13.45	13.55
42-L	10.95	11.05	38-S	15.95	16.05

Tightening Torques (MA) for Port Connections



- acc. to ISO 1179-1 / DIN 3852-T2-form X
- BSPP thread G

Series	Tube O.D.	Thread G	MA / Nm						
			Straight male stud fittings with port tapping			Non-return valves RHV/RHZ Form E with ED-sealing	Banjo fittings		Blanking plugs VSTI-ED Form E with ED-sealing
			Form A for sealing washer	Form B with cutting face	Form E with ED-sealing		WH / TH	SWVE	
L	6	G 1/8A	9	18	18	18	18	18	13
	8	G 1/4A	35	35	35	35	45	40	30
	10	G 1/4A	35	35	35	35	45	40	
	12	G 3/8A	45	70	70	50	70	65	60
	15	G 1/2A	65	140	90	85	120	90	80
	18	G 1/2A	65	100	90	65	120	90	
	22	G 3/4A	90	180	180	140	230	125	140
	28	G 1A	150	330	310	190	320		200
	35	G 1 1/4A	240	540	450	360	540		450
	42	G 1 1/2A	290	630	540	540	700		450
S	6	G 1/4A	35	55	55	45	45	40	
	8	G 1/4A	35	55	55	45	45	40	
	10	G 3/8A	45	90	80	60	70	65	
	12	G 3/8A	45	90	80	60	70	65	
	14	G 1/2A	65	150	115	145	120	90	
	16	G 1/2A	65	130	115	100	120	90	
	20	G 3/4A	90	270	180	145	230	125	
	25	G 1A	150	340	310	260	320		
	30	G 1 1/4A	240	540	450	360	540		
	38	G 1 1/2A	290	700	540	540	700		

Tolerance of tightening torques listed in above table: + 10%

Note: Lubricate stud with hydraulic oil before screwing in! Tightening torques relate to counterpart made of steel.



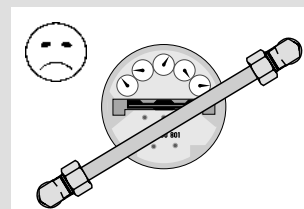
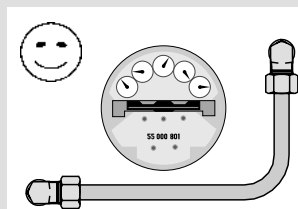
Tube Line Fabrication Guide for Leak Free Systems

Every hydraulic, pneumatic and lubrication system requires some form of tube fabrication and fitting installation for completion. Proper fabrication and installation are essential for the overall efficiency, leak free performance, and general appearance of any system.

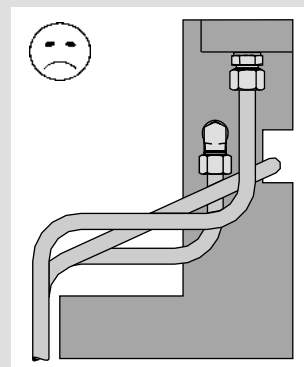
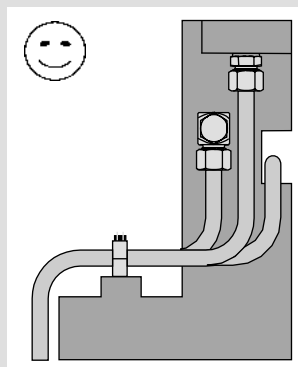
After sizing the tube lines and selecting the appropriate style of fitting, consider the following in the design of your system:

1. Accessibility of joints
2. Proper routing of lines
3. Adequate tube line supports
4. Available fabricating tools

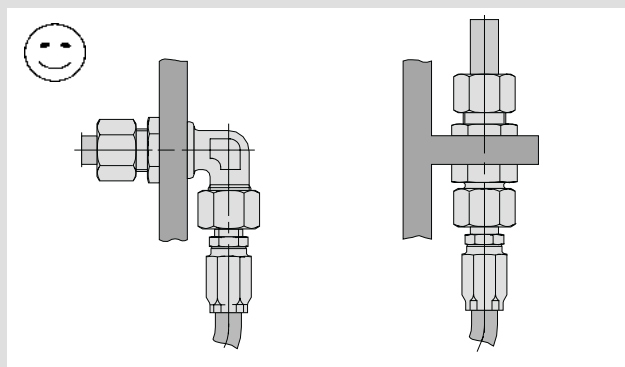
- Keep tube lines away from components that require regular maintenance:



- Have a neat appearance and allow for easy trouble-shooting, maintenance and repair:

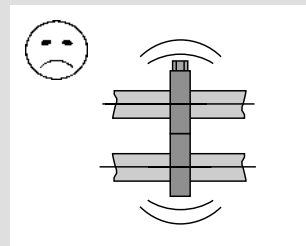
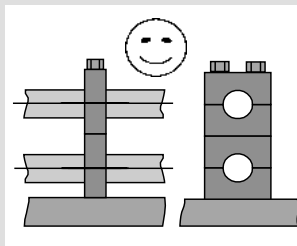


- Example for tube to hose connection:

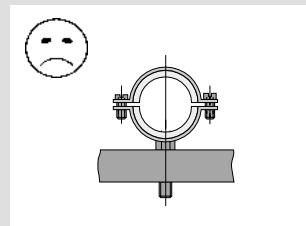
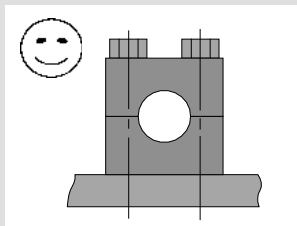


Tube Line Fabrication Guide for Leak Free Systems

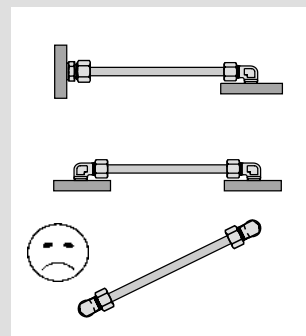
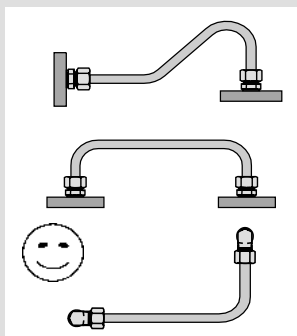
- Do not fix tubes themselves!
- Do always fix tubes on a stationary point with tube clamps:



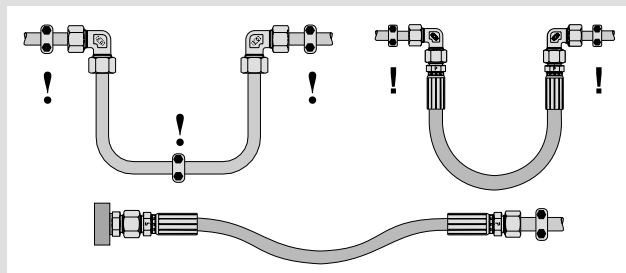
- Use appropriate tube clamps:



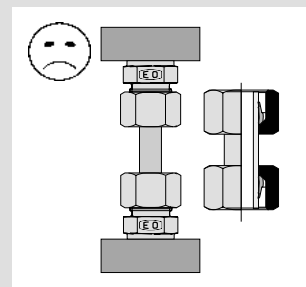
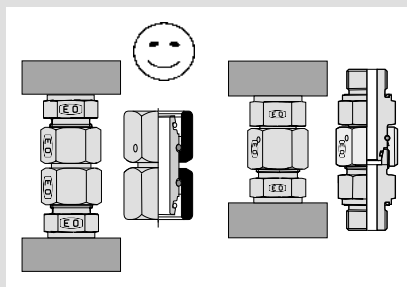
- Avoid excessive strain on joint:
A strained joint will eventually leak.



- Consider possibly length extension:

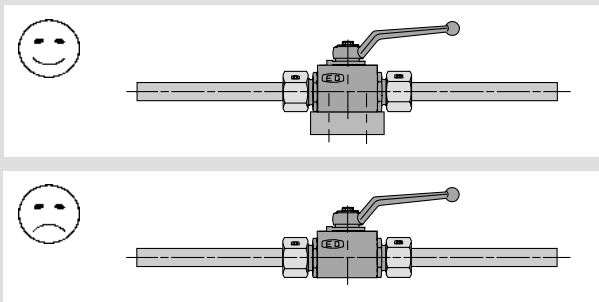


- Avoid short tube length:
- ⚠ Short tubes length are strengthen fatigue fracture.
- Use adapter GZ or swivel connector for swivel fittings instead of short tubes length.



Tube Line Fabrication Guide for Leak Free Systems

- Do cushion actuating forces:



Recommended tools for tube line fabrication:

Cutting:

EO Tube cutting tool AV

EO Combined tube bending and cutting tool BAV

Tube Cutters:

Steel: Type Kloskut;

Stainless steel: Type 635 B-EX,

Type 218 B-SS Tru-Kut Sawing Vise

Deburring:

Parker deburring tool no. 226 DEBURR.

Bending:

EO Combined tube bending and cutting tool BAV

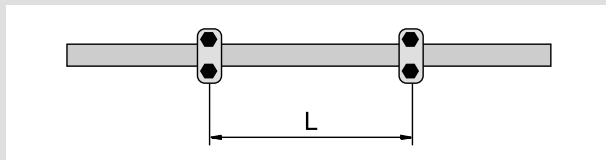
EO Tube bending tool BV 6/18, BV 20/25

EO Tube bending tool BVP (programmable)

Tube lines have to be supported in certain distances:

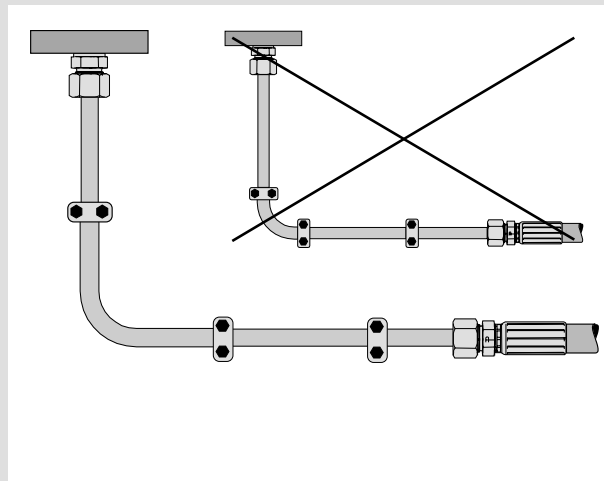
Vibration is reduced by tube clamps.

Use sufficient tube clamps.

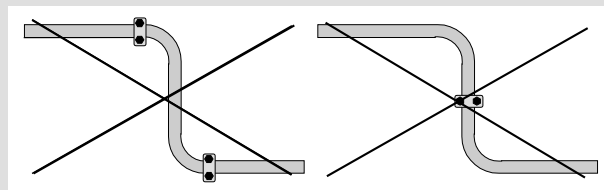
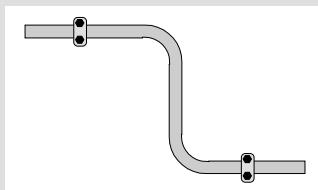


Ø [mm]	L [m]
6.0 – 12.7	1.0
12.7 – 22.0	1.2
22.0 – 32.0	1.5
32.0 – 38.0	2.0
38.0 – 57.0	2.7
57.0 – 75.0	3.0
75.0 – 76.1	3.5
76.1 – 88.9	3.7
88.9 – 102.0	4.0
102.0 – 114.0	4.5
114.0 – 168.0	5.0
168.0 – 219.0	6.0

Vibration has to be reduced near by the connectors:

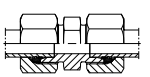
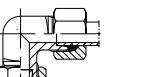
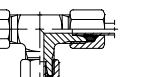
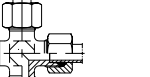
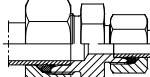
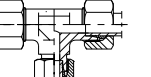
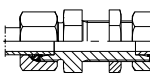
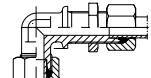
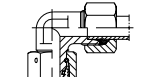
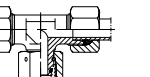
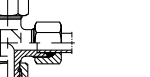
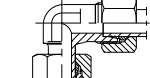
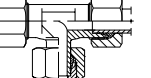
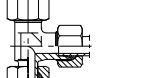
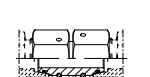
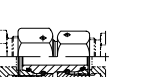
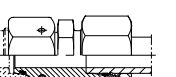
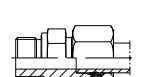
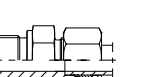
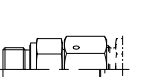
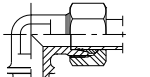
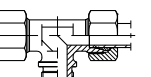
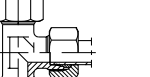


Allow expansion and contraction. Do not hamper expansion and contraction near by tube bends:

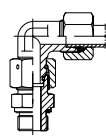
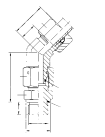
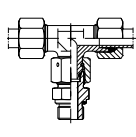
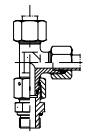
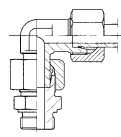
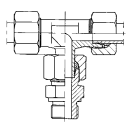
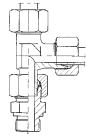
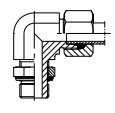
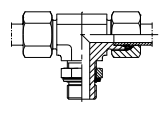
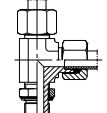
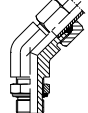
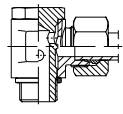
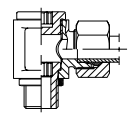
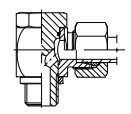
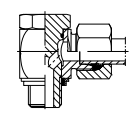
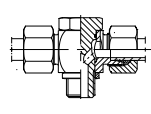
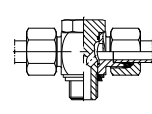
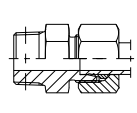
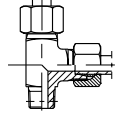
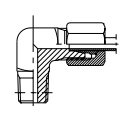
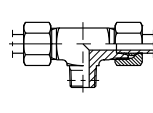
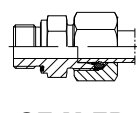
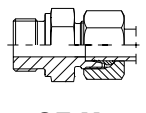
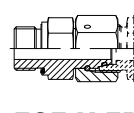
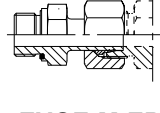
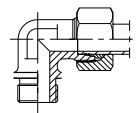
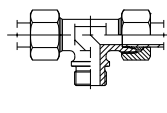
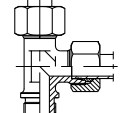
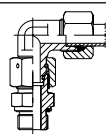
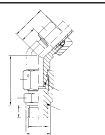
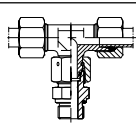
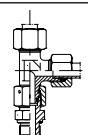
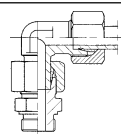
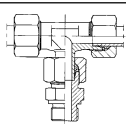
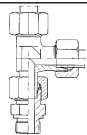
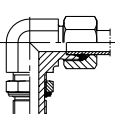
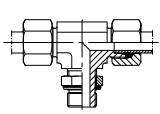
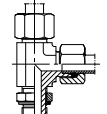
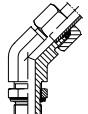


EO-2 and Progressive Ring Bite Type Fittings

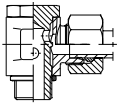
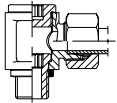
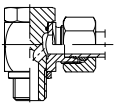
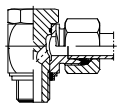
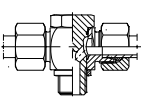
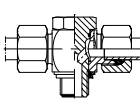
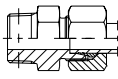
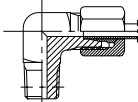
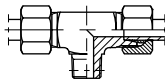
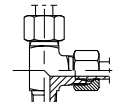
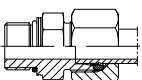
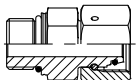
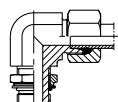
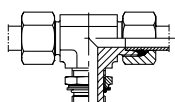
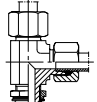
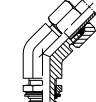
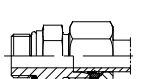
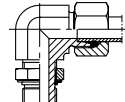
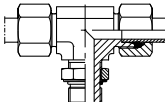
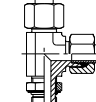
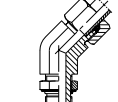
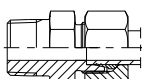
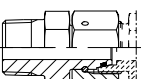
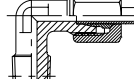
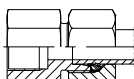
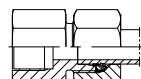
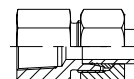
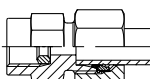
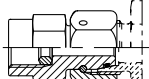
Visual Index

Tube to Tube Unions	 G p. C8	 W p. C9	 T p. C10	 K p. C11
Reducers	 GR p. C12	 TR p. C14		
Bulkhead	 SV p. C18	 WSV p. C19		
Tube to Swivel O-ring seal	 EW p. C20	 EV p. C21	 ET p. C22	 EL p. C23
Metal seal	 EVW p. C24	 EVT p. C25	 EVL p. C26	
Swivel to swivel	 GZ p. C28	 GZR p. C30	Distance Piece Adapter  DA p. C172	
Tube to BSPP Straights	 GE-R-ED p. C32	 GE-R p. C36	Swivel to BSPP  EGE-R-ED p. C40	
Non adjustables	 WE-R p. C44	 TE-R p. C45	 LE-R p. C46	

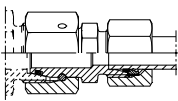
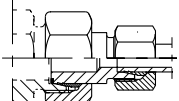
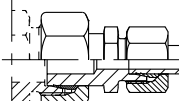
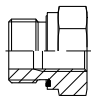
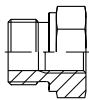
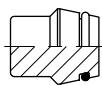
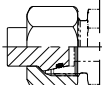
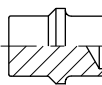
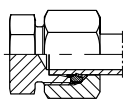
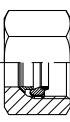
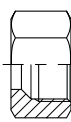
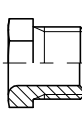
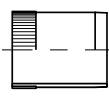
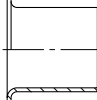
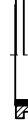
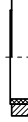
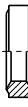
Visual Index

Swivel adjustables	 EW-R-ED p. C47	 EV-R-ED p. C48	 ET-R-ED p. C49	 EL-R-ED p. C50	 EVW-R p. C51	 EVT-R p. C52	 EVL-R p. C53
Locknut adjustables	 WEE-R p. C54	 TEE-R p. C55	 LEE-R p. C56	 VEE-R p. C57			
Banjos	 SWVE-R p. C58	 DSVW-R p. C60	 WH-R p. C62	 WH-R-KDS p. C63	 TH-R p. C64	 TH-R-KDS p. C65	
Tube to BSPT	 GE-R-KEG p. C66	 LE-R(KEG) p. C67	 WE-R(KEG) p. C68	 TE-R(KEG) p. C70			
Tube to Metric Parallel Straights	 GE-M-ED p. C72	 GE-M p. C74	Swivel to Metric Parallel		 EGE-M-ED p. C76	 EVGE-M-ED p. C78	
Non adjustables	 WE-M p. C80	 TE-M p. C81	 LE-M p. C82				
Swivel adjustables	 EW-M-ED p. C83	 EV-M-ED p. C84	 ET-M-ED p. C85	 EL-M-ED p. C86	 EVW-M p. C87	 EVT-M p. C88	 EVL-M p. C89
Locknut adjustables	 WEE-M p. C90	 TEE-M p. C91	 LEE-M p. C92	 VEE-M p. C93			

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Banjos						
	SWVE-M p. C94	DSVW-M p. C96	WH-M p. C98	WH-M-KDS p. C99	TH-M p. C100	TH-M-KDS p. C101
Tube to metric tapered						
	GE-M (KEG) p. C102	WE-M (KEG) p. C103	TE-M (KEG) p. C104	LE-M (KEG) p. C105		
Tube to metric with O-ring ISO 6149 Straights			Swivel to metric with O-ring ISO 6149			
	GEO p. C106			EGEO p. C107		
Locknut adjustables						
	WEE-OR p. 108	TEE-OR p. C109	LEE-OR p. C110	VEE-OR p. C111		
Tube to UNF						
	GE-UNF p. C112	WEE-UNF p. C114	TEE-UNF p. C115	LEE-UNF p. C116	VEE-UNF p. C117	
Tube to NPT						
	GE-NPT p. C118	EGE-NPT p. C124	WE-NPT p. C122			
Tube to female						
	GAI-R p. C126	GAI-M p. C128	GAI-NPT p. C130			
Pressure gauge adapters						
	MAV p. C132	MAVE p. C133				

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Tube end reducers	 RED p. C134	 KOR (Steel/Brass) p. C138	 KOR (Stainless Steel) p. C142		
Port reducers	 RI-ED p. C146	 RI p. C148			
Plugs Cone plugs	 VKA p. C150	 VKAM p. C151	 BUZ p. C152	 BUZM p. C153	
Port and tube plugs	 VSTI-ED p. C154	 VSTI-OR p. C155	 ROV p. C156		
Fitting components Nuts, rings and screws	 GM p. C157	 FM p. C158	 M/DPR/D p. C160	 S p. C162	
Support sleeves	 VH p. C163	 E p. C164			
Seals	 DOZ p. C165	 ED p. C166	 OR p. C166	 KDS p. C167	 KD p. C168
	 DKAZ p. C170	 DKA p. C169	 RRS p. C169	 DKI p. C166	





How to order EO-Fittings:

Fitting Material/coating

A3C	Steel, Zinc-plated and yellow chromated (standard)
71	Stainless steel
MS	Brass
VZ	Zinc-plated (Tube only)
(blank)	Steel, black phosphated (Special)

Assembly

OMD	Without Nut and Ring, but including all other components such as seals
X	Body only Basic Fitting Body without any Nut, Ring, Seal or other Component

Seal Material

NBR	Nitrile Rubber (Omitted for Steel and Brass Fittings)
VIT	Fluorocarbon FPM (Omitted for Stainless Steel Fittings)

Nomenclature

Example: **GE08ZLR1/2EDA3C**

Type

Code for shape or style see visual index

Tube-O.D.
Dia in mmExample:
08 8 mm Dia

Series

LL	Very Light
L	Light
S	Heavy

Thread Type and Size

Type

M...	Metric Parallel
M...KEG	Metric Tapered
R...	BSPP
R...KEG	BSPT
...UN/UNF	SAE Straight
...NPT	Pipe Thread

Size

No Size:	Standard Thread Size
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Example:
1/2 1/2"
(only shown on jump sizes)

Seal type

(blank)	Standard Seal
ISO 6149	O-ring
M/R	Metal Cutting Edge
M/R KEG	Tapered Thread
UN/UNF	O-ring
NPT	Tapered Thread
ED M/R	Eolastic Seal
KD WH/TH	Plastic Banjo Seal
KDS WH/TH	Elastomeric Banjo Seal

DPR/EO-2 Version

(blank)	Progressive Ring DPR or Cutting Ring D for LL-Series and Brass Fittings
Z	EO-2

Examples

Part no.	Description	Part no.	Description
GE12ZLR1/2EDA3C	Straight male Stud, EO-2, 12 mm Tube O.D., Light Series, G 1/2 BSPP, Eolastic Seal, Complete with nut and ring, zinc-plated Steel Fitting, All Seals NBR	EVT08LOMDMS	Adjustable Standpipe Branch Tee, 8 mm Tube O.D., Light Series, Brass Fitting without Nut and Ring, Standpipe preassembled with Nut and Ring.
GE12LR71X	Straight Male Stud, 12 mm Tube O.D., Light Series, G 3/8 BSPP, Metal Seal Type B, Stainless Steel Fitting, body only	EL38VITOMDA3C	Adjustable Swivel Nut Run Tee 38 mm tube O.D., Heavy Series, Zinc-plated Steel Fitting without Nut and Ring. Swivel Nut End with FPM Seal
		DOZ04LL	Spare Seal for EO-2 Joints, 4 mm O.D., Very Light Series, Steel with Nitrile Rubber Seal